J4 Position vectors

Name:

Class:

Date:

Time: **117 minutes**

Marks: **96 marks**

Comments:

Q1.

A boat moves so that its position vector, **r** metres, at time t seconds is given by

$$\mathbf{r} = (4t - 0.01t^2)\mathbf{i} + (5 - 3t - 0.04t^2)\mathbf{j}$$

where the unit vectors **i** and **j** are directed east and north respectively.

(a) State the position of the boat when $t = 0$.

(1)

(b) Find the time when the boat is due south of its initial position found in part (a).

(4)

(c) Find the time when the boat is travelling south east.

(5)

(Total 10 marks)

Q2.

A model aeroplane moves in a horizontal plane with a constant acceleration. Initially the aeroplane is at the origin and has velocity (35**i** + 45**j**) m s^{-1}. After accelerating for 8 seconds, the velocity of the aeroplane is (19**i** + 13**j**) m s^{-1}. The unit vectors **i** and **j** are perpendicular and lie in the horizontal plane.

(a) Show that the acceleration of the aeroplane is (- 2**i** - 4**j**) m s^{-2}.

(3)

(b) Find an expression for the position vector of the aeroplane at time t seconds.

(3)

(c) Find the time when the position vector of the aeroplane is (300**i** + 225**j**) m.

(7)

(Total 13 marks)

Q3.

A particle is initially at the origin. The particle has a constant acceleration of (2**i** – 3**j**) m s^{-2}.
Two seconds after it has left the origin the velocity of the particle is (2**i** + 14**j**) m s^{-1}.
The unit vectors **i** and **j** are perpendicular.

(a) Show that the initial velocity of the particle is (–2**i** + 20**j**) m s^{-1}.

(3)

(b) Find the distance of the particle from the origin after 3 seconds.

(5)

(c) The speed of the particle is 10 m s^{-1} for the first time T seconds after it has left the origin. Find T.

(6)

(Total 14 marks)

Q4.

A particle has mass 2000 kg. A single force, **F =** $1000t$**i – 5000j** newtons, acts on the particle, at time t seconds. The unit vectors **i** and **j** are perpendicular. No other forces act on the particle.

(a) Find an expression for the acceleration of the particle.

(2)

(b) At time t = 0, the velocity of the particle is 6**j** m s^{-1}. Show that at time t the velocity, **v** m s^{-1}, of the particle is given by

$$\mathbf{v} = \frac{t^2}{4}\mathbf{i} + \left(6 - \frac{5t}{2}\right)\mathbf{j}$$

(4)

(c) The particle is initially at the origin. Find an expression for the position vector, **r** metres, of the particle at time t seconds.

(4)
(Total 10 marks)

Q5.

A particle moves with a constant acceleration of (– 2**i** – 4**j**) m s^{-2}. At time $t = 0$ the particle is at the origin and has velocity (4**i** + 7**j**) m s^{-1}. The unit vectors **i** and **j** are perpendicular.

(a) Find an expression for the position vector of the particle at time t seconds.

(2)

(b) Find the time when the particle has position vector (3**i** + 5**j**) metres.

(6)
(Total 8 marks)

Q6.

A boat moves with a constant acceleration of (0.2**i** + 0.1**j**) m s^{-2}, where the unit vectors **i** and **j** are directed east and north respectively. Time t is measured in seconds.

At time t = 10, the boat is at the point A, which has position vector (30**i** – 35**j**) metres.

At time t = 20, the boat is at the point B, which has position vector (70**i** – 40**j**) metres.

(a) Find the vector $\overrightarrow{AB}$.

(1)

(b) Find the velocity of the boat when t = 10.

(3)

(c) Using your answer to part (b), find the velocity of the boat when t = 0.

(3)

(d) Find the position vector of the boat when $t = 0$.

(3)
(Total 10 marks)

Q7.

A boat moves so that its position vector, **r** metres, at time t seconds is

$$\mathbf{r} = 40\cos\left(\frac{t}{20}\right)\mathbf{i} + 80\sin\left(\frac{t}{20}\right)\mathbf{j}$$

The unit vectors **i** and **j** are directed east and north respectively.

(a) Find an expression for the velocity of the boat at time t.

(3)

(b) In what direction is the boat travelling when $t = 0$? Justify your answer.

(2)

(c) At what time is the boat travelling due south for the first time?

(2)
(Total 7 marks)

Q8.

A particle moves on a horizontal plane subject to the following three forces:

$\mathbf{F}_1 = (2\mathbf{i} + 3\mathbf{j})$N, $\mathbf{F}_2 = (6\mathbf{i} - 8\mathbf{j})$N and $\mathbf{F}_3 = (-2\mathbf{i} + 10\mathbf{j})$N

The unit vectors **i** and **j** are perpendicular and lie in the horizontal plane.

Do not include any other forces in your calculations.

(a) Find the resultant force on the particle.

(2)

(b) The mass of the particle is 8 kg. Find the acceleration of the particle.

(1)

(c) At time $t = 0$, the particle is at the origin and has velocity $(3\mathbf{i} - 2\mathbf{j})\text{ms}^{-1}$.

(i) Find the velocity of the particle when $t = 40$ seconds.

(2)

(ii) Find the distance of the particle from the origin when $t = 40$ seconds.

(5)
(Total 10 marks)

Q9.

At time $t = 0$, a particle is at the origin and moving with velocity $(4\mathbf{i} + 2\mathbf{j})$ m s^{-1}.

When t = 10 seconds the position vector of the particle is (44**i** + 28**j**) metres.
The particle is moving with constant acceleration.

(a) Find the acceleration of the particle.

(4)

(b) Find the position vector of the particle at time t.

(3)

(c) At time $t = T$, the position vector of the particle is (96**i** + 72**j**) metres.

(i) Find T.

(4)

(ii) Find the speed of the particle at this time.

(3)

(Total 14 marks)